

Security Analyst

Summary

- Monitor activity taking place on the devices and network of an organization.
- Investigate unusual or suspicious activity ie: strange logins.
- Work with other teams in a their system to improve the organization's defenses.

Responsibilities, Skills

Security analysts serve as the first responders to digital threats.

Tasks and duties include:

Continuous Monitoring: Observing SIEM dashboards 24/7/365 for anomalous activity.

Alert Triage: Separating "noise" (false positives) from genuine attacks (true positives).

Incident Investigation: Using endpoint telemetry (EDR) and network logs to reconstruct attack timelines.

Containment & Eradication: Taking technical actions like isolating infected hosts, revoking credentials, or blocking malicious IPs.

Tools

Defensive & Monitoring Tools: Management of firewalls, intrusion detection systems (IDS), and SIEM (Security Information and Event Management) platforms like Splunk or the ELK stack to monitor networks for suspicious activity.

Offensive Security Tools: For penetration testing and ethical hacking, specialists use suites such as Metasploit and Burp Suite.

Vulnerability Assessment: utilize various vulnerability-scanning and penetration-testing tools to find weaknesses before attackers do.

Sample deliverables

- - Python-Based Security "Swiss Army Knife": Custom scripts for automated reconnaissance, such as a tool that finds subdomains or scans for open ports using the Nmap library.
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- - Cryptographic Password Manager: A password manager in Python that uses a cryptography library to encrypt and securely store credentials, demonstrating your understanding of data persistence and security principles.
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- - Automated Data Quality & QA Pipeline: An ETL (extract, transform, load.) pipeline that takes raw, "messy" data (like inconsistent address lists or incident logs), geocodes it, and scores it for accuracy to reduce manual backlogs.
- - IoT Security Vulnerability Lab: Set up a home lab to test connected devices. Analyze firmware for hardcoded credentials or unencrypted communication protocols, then document how you would harden the device.

My ideal workplace would be at large tech job with a structured work style, I don't mind working independently or with a team. Mentorship and freedom would be preferable. I say this because I'm changing careers and starting fresh, from information I've gathered, big tech offers high liquid pay, job security, and scale, while startups offer rapid growth, direct impact, uncertain equity, more fast pace work environments, and learn by doing. That said I would definitely consider working for a startup when I am more seasoned but as a beginner I believe big tech would be a better launch point.

